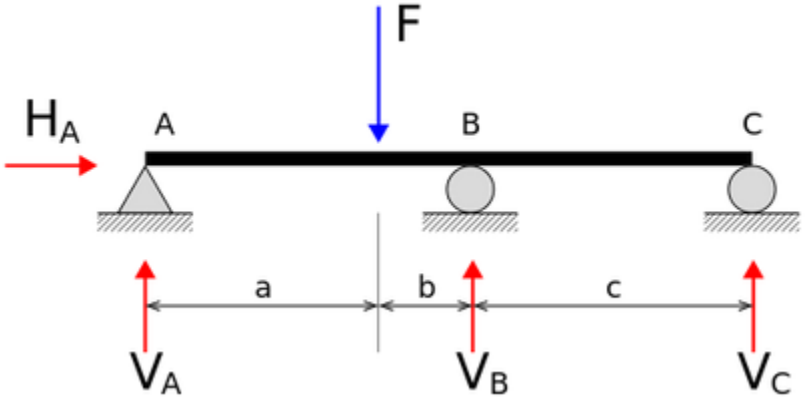
Engineering Mechanics I

Week 4

Yuxia Yuan

Autonomous Aerial Systems

Statically determinate vs indeterminate



A structure is Statically determined

→ if static equilibrium equations are sufficient to determine internal forces and reactions

↔ if and only if criterions

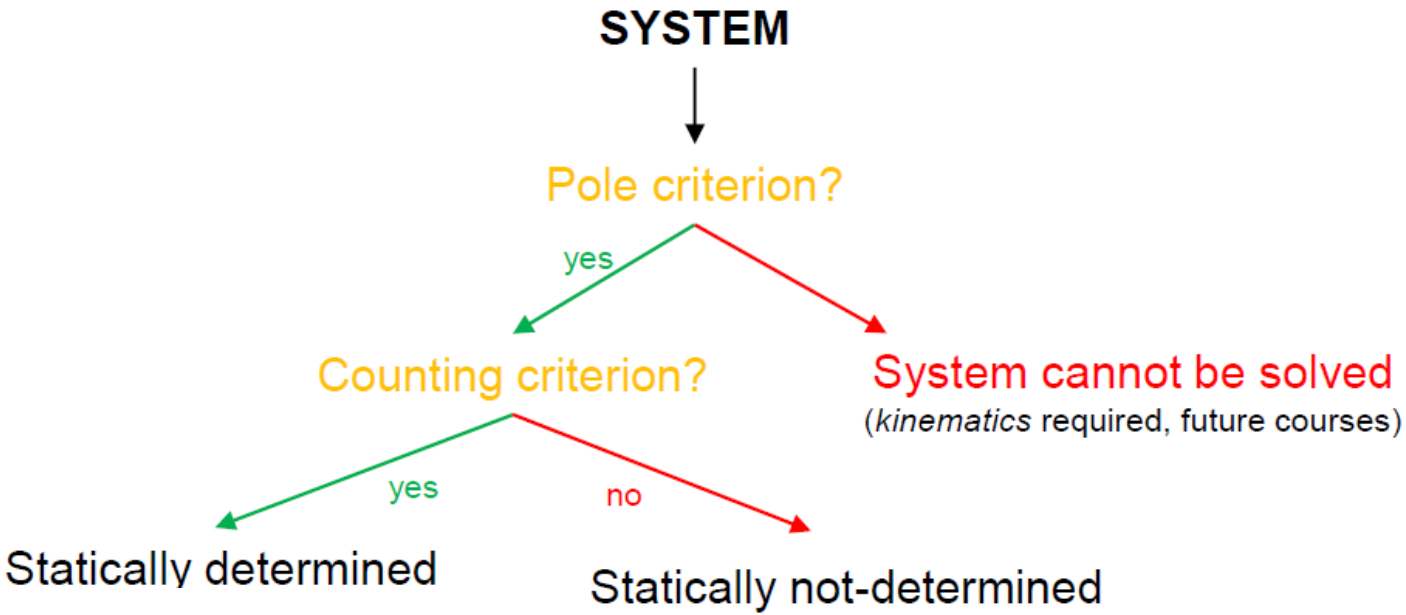
- counting criterion
- and
- pole criterion

are fulfilled

if we have enough equation

if the system is movable.

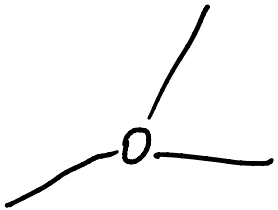
What does statically determined mean?



Counting criterion

2D (3 DOF): $S + I = 3E$

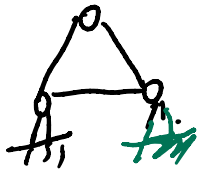
3D (6 DOF): $S + I = 6E$



with

S → # **S**upport reactions
I → # **I**ntermediate reactions
E → # **E**lements

→ rods { 2D: $2(n-1)$; 3D: $3(n-1)$



$S = 3 + 1$
$I = 6.$

$3E = S + I$ → ✗
$\neq$

Notice that

necessary but not sufficient.

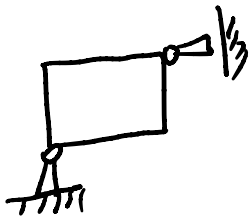
Pole-criterion

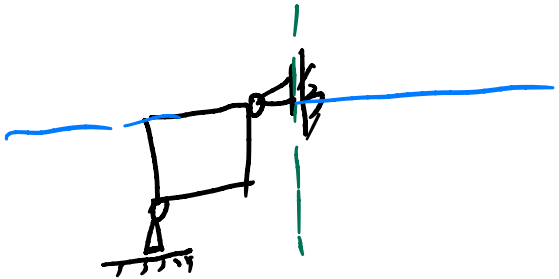
Pole-criterion:

Accounts for the displacability of the system and is fulfilled if main pole cannot be found for any structral elements

Fulfillment of the pole conditions proves that the syste is not movable.

Examples:



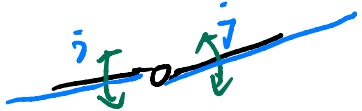


Notice that

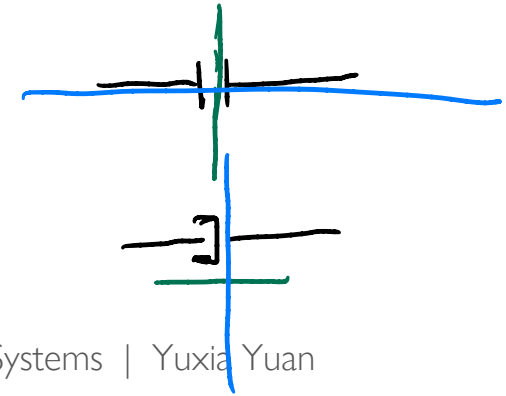
Pole-plans

- The velocity vector in planar motion is always perpendicular to a rotation line

- Pole
  - Main pole: Point around which solid i rotates or translates relative to fixed Env
    (O, j)
  - Relative pole: Point around which solid i rotates or translates relative to solid j
    (i, j)
  - Moment joint (rod) :

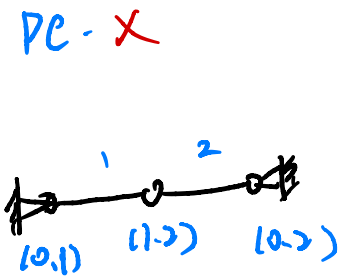


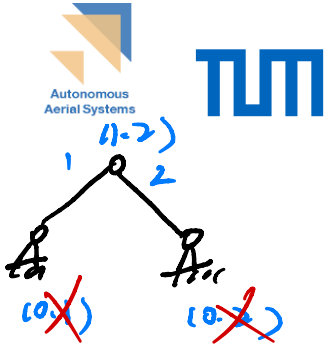
  - Tangent force joint:



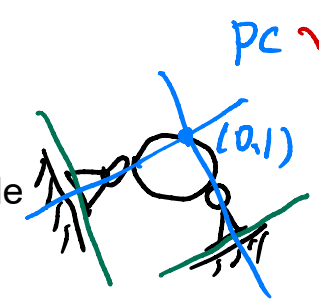
  - Normal force joint:

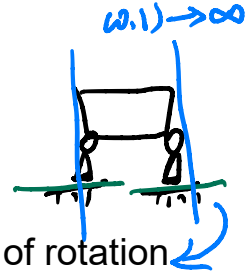
Pole-plans - Rules





- Two main poles and the respective relative pole lie in a line
- Fixed point is a geometrical location for a pole
- The line perpendicular to the motion of a joint is a geometrical location for a pole
- Parallel geometrical location lines intercept in the infinite
- A pole located in the infinite results in a translation instead of rotation

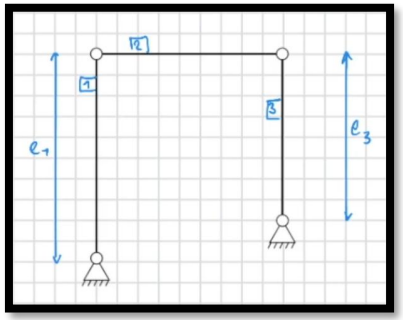


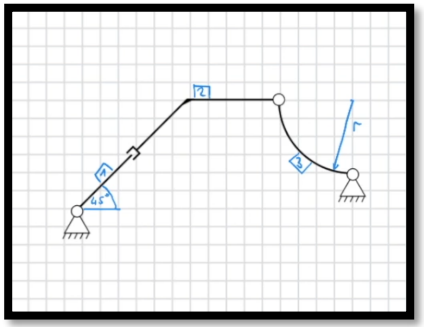


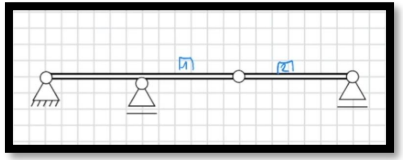
Exercises definition

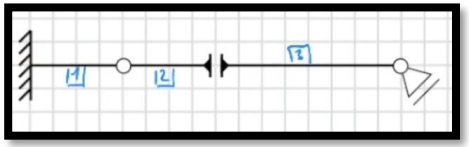
We are given…

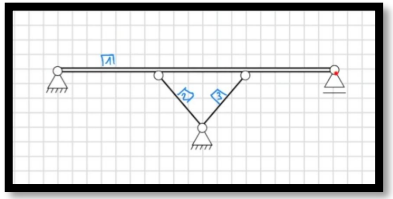
- System structure











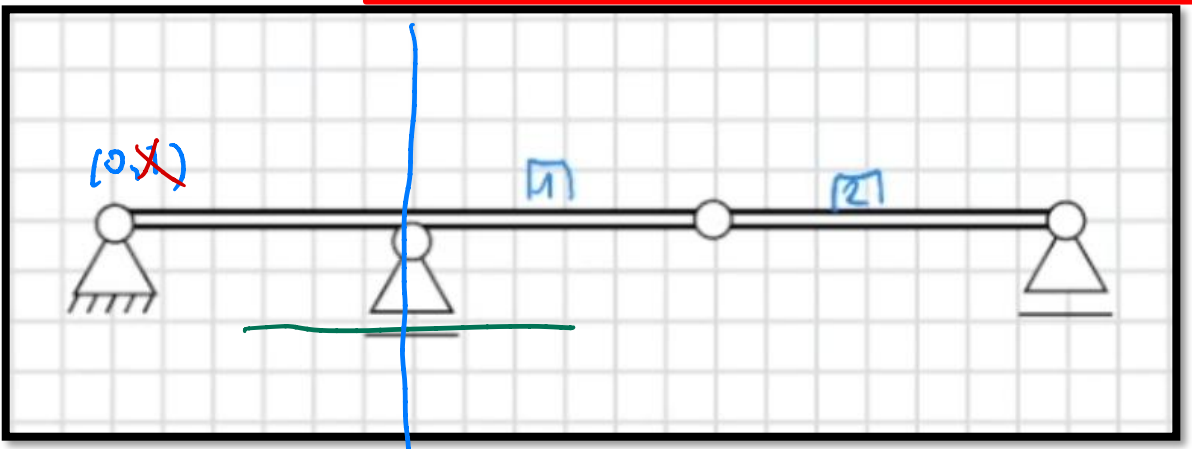
Calculate …

If the system is statically determinate

- Counting criterion
- Pole criterion
- Displacement diagram

Exercise 1

Calculate: *If the system is statically determinate with* counting criterion (I), pole criterion (II) and displacement diagram (III)



CC: ✓

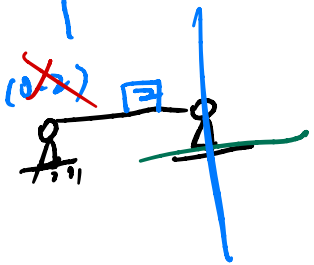
$s = 4$

$I = 2$

$s + I = 3 \cdot 2$

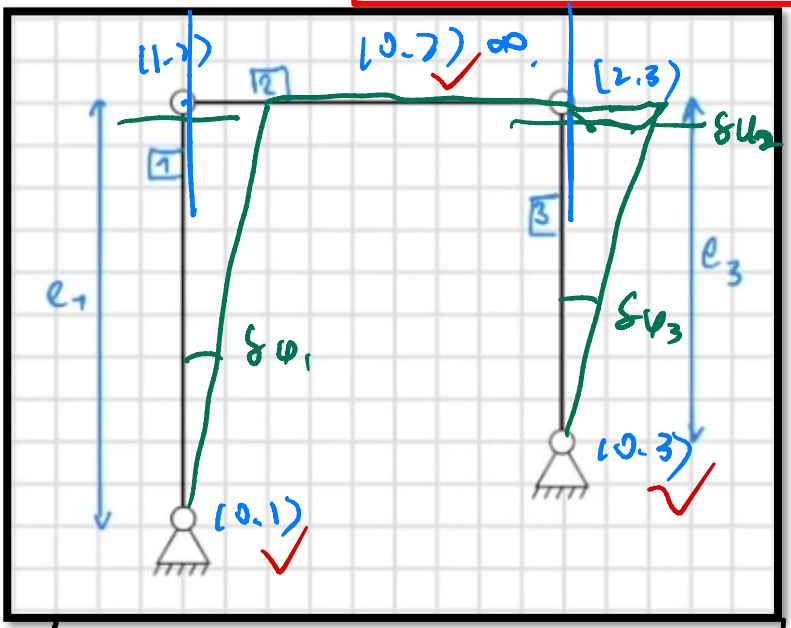
PC: ✓

⇒



Exercise 2

Calculate: *If the system is statically determinate with* counting criterion (I), pole criterion (II) and displacement diagram (III)



Small angel assumption

$\sin\theta \underset{\theta\to 0}{=} \theta$, $\cos\theta \underset{\theta\to 0}{=} 1$, $\tan\theta \underset{\theta\to 0}{=} \theta$

CC: $s + I = 3E$ ✗

$s = 4$, $I = 4$, $s + I \neq 3E$

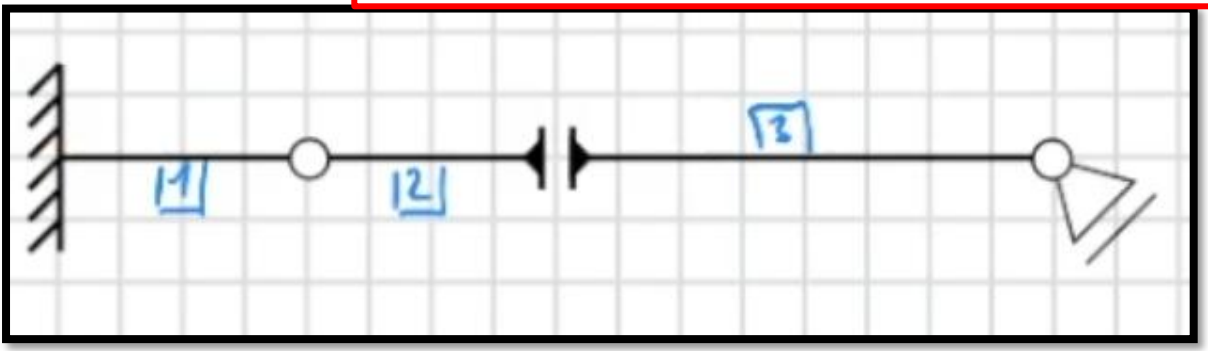
PC: ✗

→ $\delta u_2 = l_3 \cdot \sin\delta\varphi_3$

$\delta u_2 = L_1 \cdot \sin\delta\varphi_1$

Exercise 3

Calculate: *If the system is statically determinate with* counting criterion (I), pole criterion (II) and displacement diagram (III)



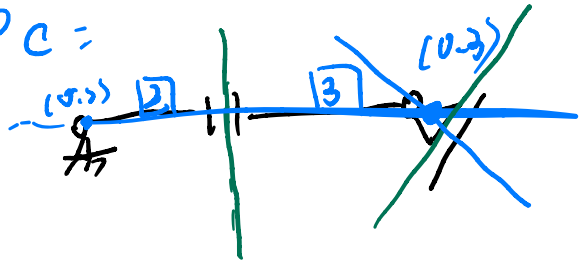
cc: ✗

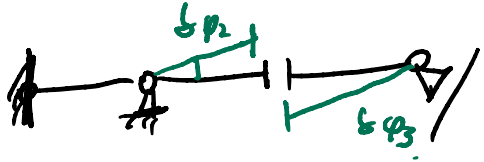
$s = 3 + 1$

$z = 2 + 2 = 4$

$s + z \neq 3E$

PC:

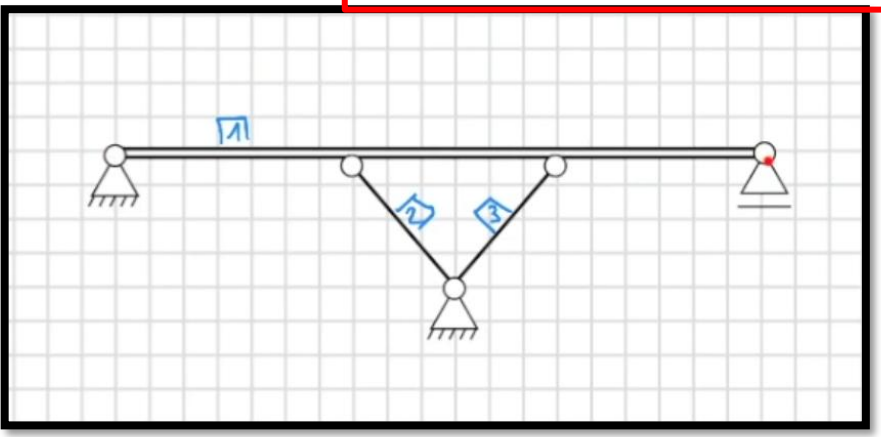




$\delta\varphi_2 = \delta\varphi_3$

Exercise 4

Calculate: *If the system is statically determinate with* counting criterion (I), pole criterion (II) and displacement diagram (III)

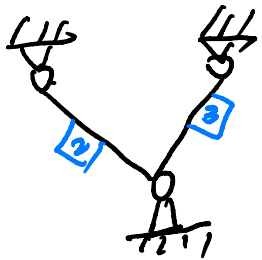


CC: ✗

$S = 2+2+1 = 5$. $Z = 2+2+2$

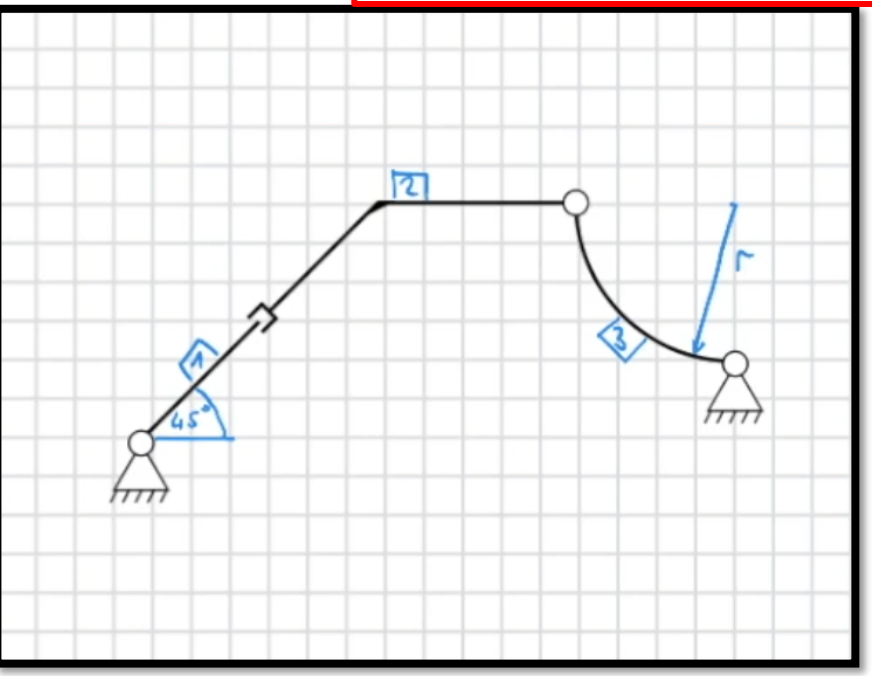
$S+Z \neq 3E$

PC ✓



Exercise 5

Calculate: *If the system is statically determinate with* counting criterion (I), pole criterion (II) and displacement diagram (III)



Engineering Mechanics I

Week 4

Yuxia Yuan

Autonomous Aerial Systems